Day 1: Introduction to Network Protocols

Report: Role of DNS and HTTP in Web Communication

When we type a website name like www.spotify.com in the browser, two protocols do the main work: DNS finds where the website is, and HTTP (or HTTPS) gets the web page from it.

DNS (Domain Name System): Computers talk to each other with IP addresses, but people remember names. DNS works like the phonebook of the internet and converts a domain name into an IP address. The browser sends a DNS query (usually over UDP port 53) to the DNS server, and the server replies with the IP address.

HTTP / HTTPS (Hypertext Transfer Protocol): After the browser knows the IP address, it connects to the web server and sends an HTTP request such as GET. The server replies with a response (for example 200 OK) that contains the page, images and data. HTTPS is the secure version of HTTP, which uses TLS encryption (port 443).

Steps of web communication:

1. The user types www.spotify.com in the browser.
2. The browser checks its cache. If the IP is not found, it sends a DNS query to the DNS server.
3. The DNS server sends back a DNS response with the IP address.
4. The browser opens a connection (TCP handshake, and the TLS handshake for HTTPS) to that IP address.
5. The browser sends an HTTP/HTTPS request, and the server sends back the web page.
6. The browser shows the page to the user.

So without DNS the browser would not know where to go, and without HTTP/HTTPS it could not get the page. They work together on every website visit.

Other protocols in this lesson (quick overview):

Protocol	Role
DNS	Converts domain names into IP addresses.
DHCP	Automatically gives an IP address, subnet mask, gateway and DNS server to a device when it joins a network.
HTTP / HTTPS	Transfers web pages and data between browser and server; HTTPS adds encryption.
FTP	Transfers files between computers (upload and download), on port 21.
SNMP	Lets a network admin monitor and manage devices like routers and switches.

Task 1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

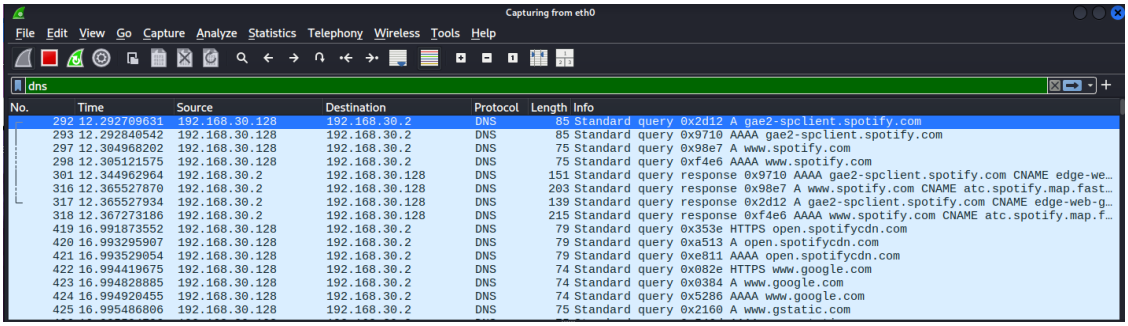
Answer:

1. Open Wireshark and start capturing on your active interface (for example eth0 or Wi-Fi).
2. Open www.spotify.com in your browser and wait until the page loads. Then stop the capture.

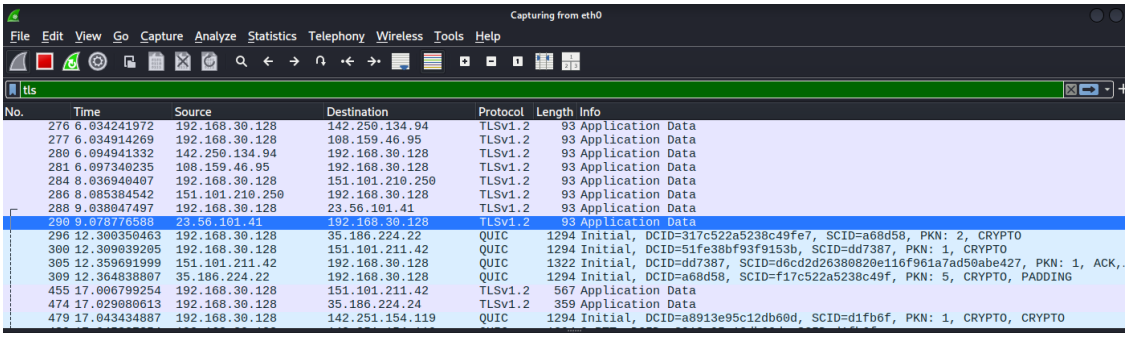
3. Type dns in the filter bar and press Enter. Note two DNS packets: a "Standard query ... www.spotify.com" and its "Standard query response".
4. Change the filter to tls (or http) and note two packets, for example "Client Hello (SNI=...spotify...)" and "Server Hello".

Tip: If you do not see any DNS packet, the browser may have the address in its cache (or it uses DNS over HTTPS). Run nslookup www.spotify.com in the terminal while capturing, and the DNS packets will appear.

No.	Packet number	Protocol	Source > Destination	Info column (what I saw)
1	297	DNS	192.168.30.128 > 192.168.30.2	Standard query 0x98e7 A www.spotify.com
2	316	DNS	192.168.30.2 > 192.168.30.128	Standard query response 0x98e7 A www.spotify.com CNAME atc.spotify.map.fastly.net ...
3	288	TLSv1.2 (HTTPS)	192.168.30.128 > 23.56.101.41	Application Data (93 bytes)
4	290	TLSv1.2 (HTTPS)	23.56.101.41 > 192.168.30.128	Application Data (93 bytes)



Screenshot 1: Wireshark with the filter "dns" while loading www.spotify.com



Screenshot 2: Wireshark with the filter "tls" (HTTPS traffic) while loading www.spotify.com

Explanation: Packet 297 is the DNS query from my computer (192.168.30.128) to the DNS server (192.168.30.2) asking for the IPv4 address (record A) of www.spotify.com. Packet 316 is the DNS server's response with the same ID (0x98e7). It shows that www.spotify.com is an alias (CNAME) of atc.spotify.map.fastly.net, a Fastly CDN server, and gives its IP address. Packets 288 and 290 are HTTPS packets (TLSv1.2): my computer sends encrypted application data and the server (23.56.101.41) replies. Wireshark shows only "Application Data" because HTTPS data is encrypted. Newer connections also use QUIC (for example packet 296), which is the transport used by HTTP/3 over UDP.

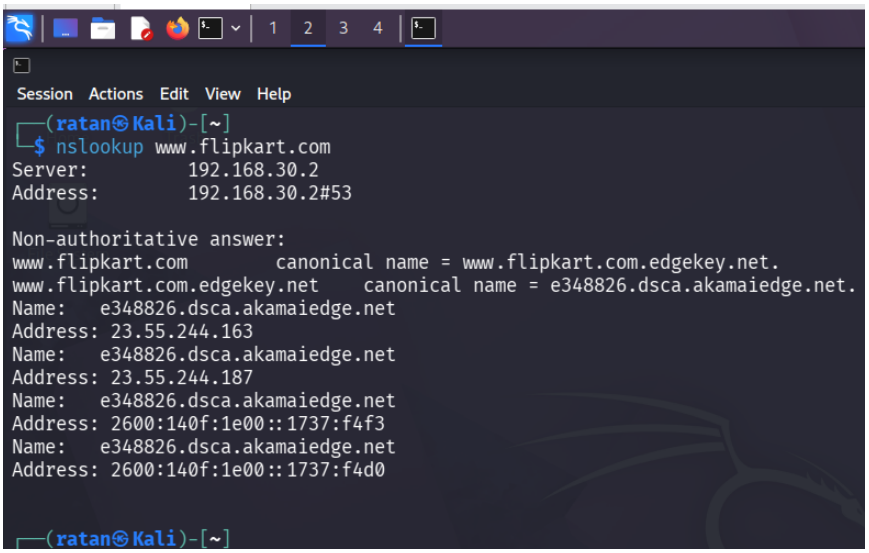
Task 2. Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

Answer:

nslookup www.flipkart.com

In the output, the first "Server" and "Address" lines show the DNS server used. The lines after "Name: www.flipkart.com" show the answer: the "Address" (or "Addresses") is the IP address of Flipkart.

IP address returned for www.flipkart.com: 23.55.244.163 (the first IPv4 address). Other addresses returned: 23.55.244.187 (IPv4), and 2600:140f:1e00::1737:f4f3 and 2600:140f:1e00::1737:f4d0 (IPv6).



Screenshot: nslookup www.flipkart.com

What this result means:

The result shows that my DNS server (192.168.30.2, port 53) converted the name www.flipkart.com into IP addresses, so my computer can now connect to Flipkart using one of them. The answer has these parts: (1) "Non-authoritative answer" means the reply came from the DNS server's cache and not directly from the official DNS server of Flipkart. (2) www.flipkart.com is only an alias: its canonical name (CNAME) is www.flipkart.com.edgekey.net, which points to e348826.dsca.akamaiedge.net. This shows that Flipkart uses the Akamai CDN, which serves the site from servers close to the user. (3) Several addresses are returned (two IPv4 and two IPv6), which means the traffic is shared between many servers, and the browser can connect to any of them. The IP addresses can be different from different places or at different times.

Task 3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

Answer:

No.	Point	HTTP	HTTPS
1	Security / encryption	Data is sent as plain text, so anyone on the network (for example on public Wi-Fi) can read it.	Data is encrypted with SSL/TLS, so it cannot be read or changed on the way.

No.	Point	HTTP	HTTPS
2	Port number	Uses port 80.	Uses port 443.
3	Certificate and identity	No certificate, so the browser cannot confirm that the website is real.	Uses an SSL/TLS certificate from a trusted authority, which proves the website identity. The browser shows a padlock icon, and the URL starts with https://.

Real-world example: Paytm (and any payment page, such as Flipkart checkout). When I enter my UPI PIN, card details or OTP, the data goes over HTTPS, so it is encrypted and no one in between can steal it. The same is true for Instagram login and Gmail. The padlock in the browser shows that the connection is secure.

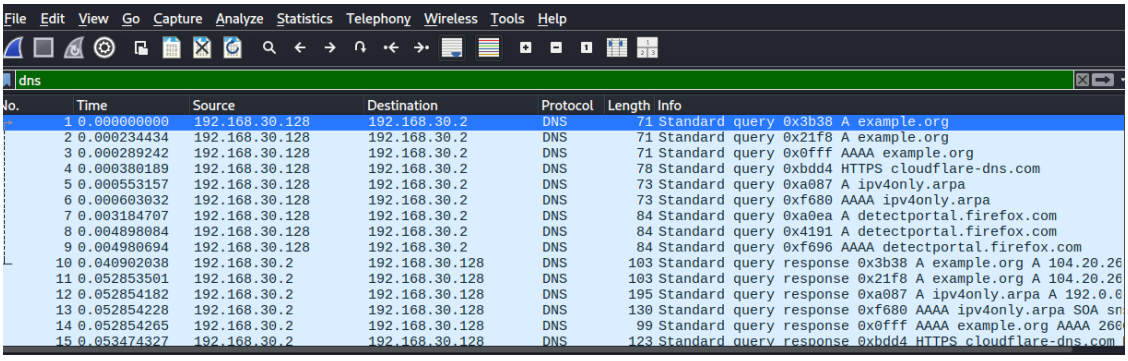
Task 4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.

Answer:

1. Start a Wireshark capture and type dns in the filter bar.
2. In the browser open a website you have not visited recently (write its name below).
3. Look at the packets in the order of the Time column and note each "Standard query" and its "Standard query response".

Website I opened: example.org (a site I had not visited recently)

No.	Time (s)	Source > Destination	Type	Info column (what I saw)
1	0.000000	My PC > DNS server	Standard query	0x3b38 A example.org
2	0.000234	My PC > DNS server	Standard query	0x21f8 A example.org (query repeated)
3	0.000289	My PC > DNS server	Standard query	0x0fff AAAA example.org
10	0.040902	DNS server > My PC	Standard query response	0x3b38 A example.org A 104.20.26...
11	0.052853	DNS server > My PC	Standard query response	0x21f8 A example.org A 104.20.26...
14	0.052854	DNS server > My PC	Standard query response	0x0fff AAAA example.org AAAA 2606:...



Screenshot: Wireshark with the filter "dns" while opening example.org

Description of what I saw: My computer (192.168.30.128) sent the first queries to the DNS server 192.168.30.2 at time 0.000 s. It asked for the IPv4 address (A record) and the IPv6 address (AAAA record) of example.org. Only about 0.04 to 0.05 seconds later, the DNS server sent back "Standard query response" packets, each with the same ID as its query (0x3b38, 0x21f8 and 0x0fff), and they gave the IP address (for example 104.20.26.x for IPv4 and an IPv6 address for AAAA). Other queries appeared at the same time from the browser, such as detectportal.firefox.com (a connectivity check) and cloudflare-dns.com. After the responses, the browser knows the IP address and can connect to the website.

General sequence (what normally happens):

1. The browser sends a "Standard query A <website>" to the DNS server to ask for the IPv4 address. Often a second query, "AAAA", is sent at the same time for the IPv6 address.
2. The DNS server sends a "Standard query response" for each query. It contains the IP address (A or AAAA record), and sometimes an alias (CNAME) that points to another name, for example a CDN name, followed by its IP address.
3. Each response has the same Transaction ID as its query, which is how Wireshark and the computer match a reply to its question.
4. After the response, the browser connects to the IP address (TCP / TLS packets begin), and the page loads. If the site uses other domains (images, ads, scripts), more DNS queries and responses appear.

Task 5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

Answer:

Protocol chosen: DHCP (Dynamic Host Configuration Protocol)

Scenario: I connect my laptop to my phone's Wi-Fi hotspot and then open YouTube. When the laptop joins the Wi-Fi, it does not have any IP address, so it uses DHCP in four steps (DORA):

1. Discover: the laptop broadcasts a message asking "Is there a DHCP server?".
2. Offer: the hotspot / router, which works as the DHCP server, replies with an offer of an IP address.
3. Request: the laptop says "I accept this address".
4. Acknowledge: the DHCP server confirms and gives the details for a limited time (a lease).

In my own laptop, ipconfig /all showed that the DHCP server was 10.123.18.168, which gave my laptop the IP address 10.123.18.212, subnet mask 255.255.255.0, default gateway 10.123.18.168 and DNS server 10.123.18.168. Because of this, I did not have to type any network settings by hand. After that, my laptop used DNS to find the YouTube server and HTTPS to stream the video. Without DHCP, I would have to enter the IP address, mask, gateway and DNS manually every time I join a new Wi-Fi, and two devices could easily get the same IP address by mistake.